WATER PUMP Dry-Run Guard

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Here is an effective solution for protection of household submersible water pumps against dry running. Minimum water-level monitoring feature in the circuit is realised using suspended sensor probes to ensure that the water pumps will not run under a dry condition. Besides, switch-on and return-delay features prevent the pumps from unwanted brief operation when handling turbulent underground water. Fig. 1 shows the author's prototype.

Fig. 1(a): Front panel of the author's prototype

Circuit and working

Circuit diagram of the water pump dry-run guard is shown in Fig. 2. It is built around 5V regulator 7805 (IC1), timer NE555 (IC2), transistors BC547 (T1) and SL100 (T2), a 12V 1C/O relay (RL1) and a few other components.

To understand how this circuit works, assume that output pin 3 of IC2 is high during initial power up and, hence, the relay RL1 is in energised state. In this case, capacitor C4 at the input of IC2 will be charged via resistor R2 and potmeter VR2.

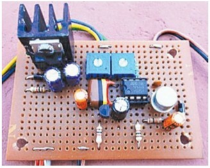

Fig. 1(b): Author's prototype wired on a perfboard

After a few seconds delay, voltage on C4 will reach a level that causes the inverter circuit (formed by IC2) to change state. IC2 output

PARTS LIST

Semiconductors:	
IC1	- 7805, 5V voltage regulator
IC2	- NE555 timer
T1	- BC547 npn transistor
T2	- SL100 npn transistor
LED1, LED2	- 5mm LED
D1, D2	- 1N4007 rectifier diode
Resistors (all 1/4-watt, ±5% carbon):	
R1, R3-R5	- 1-kilo-ohm
R2	- 100-kilo-ohm
VR1	- 2.2-mega-ohm potmeter
VR2	- 470-kilo-ohm potmeter
Capacitors:	
C1	- 100µF, 25V electrolytic
C2	- 100µF, 16V electrolytic
C3, C4, C7	- 10µF, 16V electrolytic
C5	- 0.1µF ceramic disk
C6	- 0.01µ ceramic disk
Miscellaneous:	
CON1, CON2	- 2-pin terminal connector
S1	- Tactile switch
RL1	- 12V, 1C/O, 30A electromagnetic relay
	- 1.5HP submersible bore-well pump
	- 12V, 1A regulated power supply
AS1, AS2	- Stainless steel bolt

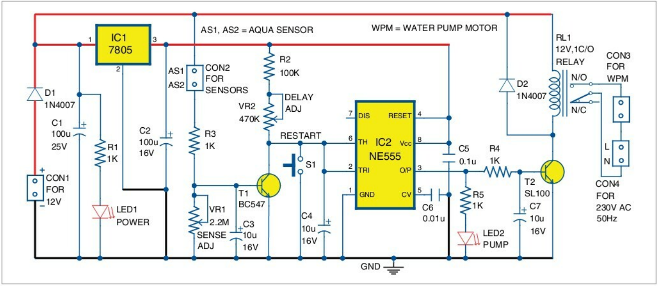

Fig. 2: Circuit diagram of the water pump dry-run guard